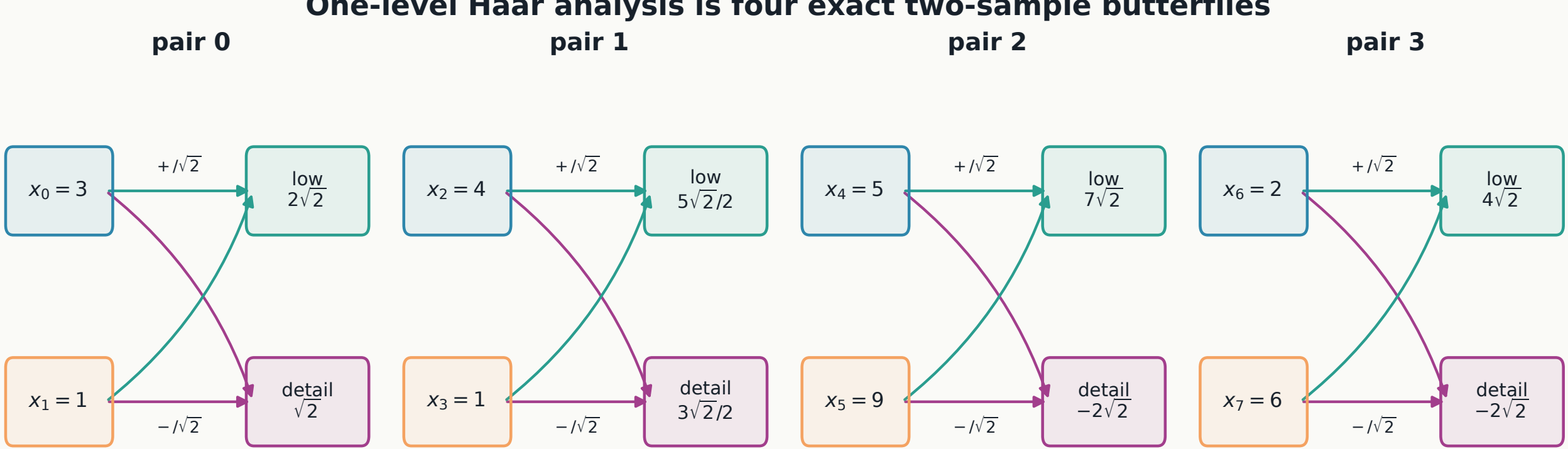
One-level Haar analysis is four exact two-sample butterflies
pair 0
pair 1
pair 2
pair 3
$x_0 = 3$
$x_1 = 1$
$+/\sqrt{2}$
$-/\sqrt{2}$
low $2\sqrt{2}$
detail $\sqrt{2}$
$x_2 = 4$
$x_3 = 1$
$+/\sqrt{2}$
$-/\sqrt{2}$
low $5\sqrt{2}/2$
detail $3\sqrt{2}/2$
$x_4 = 5$
$x_5 = 9$
$+/\sqrt{2}$
$-/\sqrt{2}$
low $7\sqrt{2}$
detail $-2\sqrt{2}$
$x_6 = 2$
$x_7 = 6$
$+/\sqrt{2}$
$-/\sqrt{2}$
low $4\sqrt{2}$
detail $-2\sqrt{2}$